

Ghanashyam Bharadwaz Avula

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SUMMARY

Computer Engineering graduate with experience in building production grade AI applications, combining software engineering with LLMs to deliver scalable, real-world solutions. Take a closer look at my work in my [Portfolio](#). Looking for backend or AI engineering roles where I can ship production systems. Certified by Anthropic and IBM in Generative AI and machine learning.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

Frameworks: React, Node.js, FastAPI

Databases: PostgreSQL, MongoDB, Redis

DevOps & Cloud: Git, Docker, AWS, GCP

Gen AI: LangChain, Agentic AI, RAG, MCP, LLM Orchestration, LoRA

ML: PyTorch, TensorFlow, Scikit-learn, Seaborn

PROJECTS

Ecom Support Assistant | *Python, FastAPI, MCP, RAG, PostgreSQL, Evals*

[\[Live Link\]](#)

- Deployed a production grade e-commerce customer support assistant that answers queries grounded in store policy docs and a live orders database, exposed as an MCP server with a glass-box UI that streams the tool-call trace in real time.
- Designed a custom Model Context Protocol (MCP) architecture integrating multiple tools like RAG, order status retrieval and summarization with coordinated orchestration for seamless multi-tool execution.
- Validated on a 34-case eval harness of answerable, adversarial, and undocumented-topic queries: 33/34 answers fully grounded in retrieved context (LLM as judge), 97% retrieval recall and every out-of-scope query declined with zero false refusals.

AskMyDB - NL to SQL Engine | *LangChain, Node.js, Llama LLM, Graph RAG, SQLite, LoRA*

[\[Live Link\]](#)

- Orchestrated a natural language to SQL pipeline with hybrid retrieval (Vector + Graph RAG) for inter-table relations, multi-layer safety guardrails, and zero-config database onboarding for plain English to optimized SQL conversion.
- Engineered a LangChain pipeline using Llama LLM with Node.js middleware for JWT authentication, conversation memory, and robust guardrails for write-operation interception, rate limiting, and destructive query prevention, with 100% success across 31 functional test cases including complex multi-table joins.
- Evaluated the performance of LoRA finetuned Llama 8B model, which cut latency by 83% and cost by 92% vs. the 70B baseline model on a fixed schema, but kept the frontier model as default after it performed better on unseen databases.

Distributed Rate Limiter | *REST API, FastAPI, Redis, Lua, Docker, Nginx*

[\[Github\]](#)

- Built a Redis-backed distributed rate-limiting middleware for FastAPI that enforces exact per-key request limits across multiple load-balanced replicas, with a live dashboard to fire request bursts and watch limiter decisions in real time.
- Implemented a token bucket algorithm as an atomic Redis Lua script, eliminating read-check-write race conditions, with configurable fail-open/fail-closed modes on a Dockerized Nginx and 3-replica stack.
- Proved exactness under load with a zero-tolerance concurrency test admitting exactly 100 of 2,000 concurrent requests across replicas, adding only ~0.8ms latency overhead while sustaining 4,000+ req/s throughput via Nginx.

EXPERIENCE

AICTE Google AI-ML Virtual Internship

June 2024 – August 2024

Google / AICTE

Remote

- Worked on integrating multiple image classification models using TensorFlow and Keras for ‘Breast cancer detection’.
- Improved F1 score from 0.78 to 0.92 on a dataset containing over 2000 images.

EDUCATION

Sreenidhi Institute of Science and Technology (Hyderabad, IN)

Bachelor of Technology in Computer Engineering

2022 – 2026

ACHIEVEMENTS & CERTIFICATIONS

- Participated in Smart India Hackathon 2023 as a backend developer, collaborating with my team on “Smart Crop Management System for Farmers” project.
- Claude code using Google Cloud Vertex AI– Anthropic
- IBM Machine Learning Specialization – Coursera
- Secured 97 percentile in GATE 2026.